PHASE 2 · EXPERIMENTAL RESULTS & COMPARISON

Evidence-Gated Natural-Language Explainable AI for Malaria Detection

A comparison of the published study, our XAI implementation, and the Phase 2 reporting extension

1. Study and comparison scope

The reference study by Parveen et al. (2025) investigates deep-learning-based malaria classification using Xception and Inception-ResNetV2, with Grad-CAM, LIME and SHAP to explain model predictions. Our project implements the two model pipelines and explanation methods, then adds a Phase 2 reporting layer that checks whether explanation evidence and limitations are stated consistently.

The comparison has three distinct parts: (i) published classification results versus the scope of our own pilot; (ii) qualitative visual comparison of the paper's Xception examples with our supplied Xception outputs; and (iii) before-and-after coverage of evidence and limitation fields in the reporting layer.

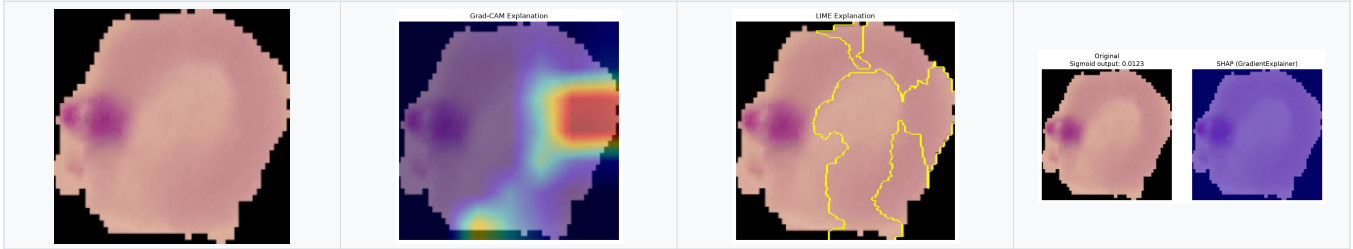
2. Results reported in the original paper

The paper reports the following validation and test accuracies on its two datasets. These are published results from the authors' experiments, not results from our pilot.

Architecture / dataset	Validation accuracy	Test accuracy
Xception · Kaggle	97.7%	97.6%
Inception-ResNetV2 · Kaggle	97.8%	97.8%
Xception · Harvard Dataverse	97.7%	97.5%
Inception-ResNetV2 · Harvard Dataverse	97.6%	97.6%

Important distinction: Our 20-image pilot (10 Parasitized and 10 Uninfected) produced 20/20 correct predictions. This is a small exploratory sample, not a like-for-like reproduction of the paper's dataset-level test evaluation. It must not be presented as 100% test accuracy or as a direct improvement over the published results.

3. Our implementation at a glance



Input image	Grad-CAM	LIME	SHAP

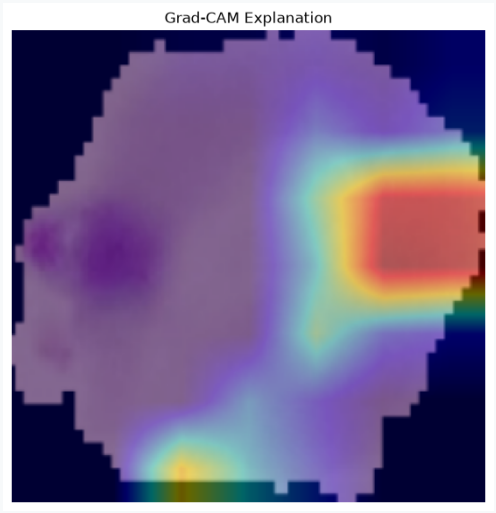
The supplied implementation image displays a sigmoid output of 0.0123. The class label associated with this value depends on the model's class-index mapping; the value is therefore reported without independently assigning a class here.

VISUAL COMPARISON · XCEPTION

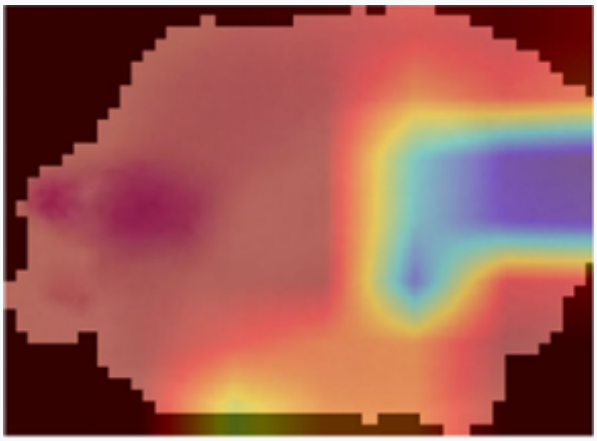
What the explanation maps show

The following pairs compare our supplied outputs with the Xception examples from the original paper. The comparison is visual and qualitative; the paper's examples are not confirmed to use the exact same input cell as ours.

4. Grad-CAM: similar broad focus, different map shape



Our implementation



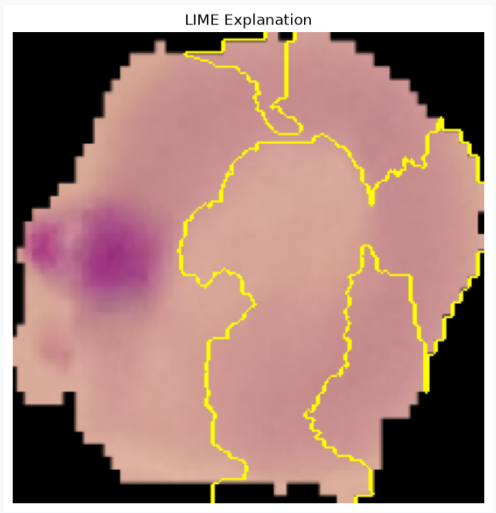
Original paper · Xception

Similarity. Both examples place a strong activation area on the right-hand side of the cell, suggesting a broadly comparable region of model attention in these particular images.

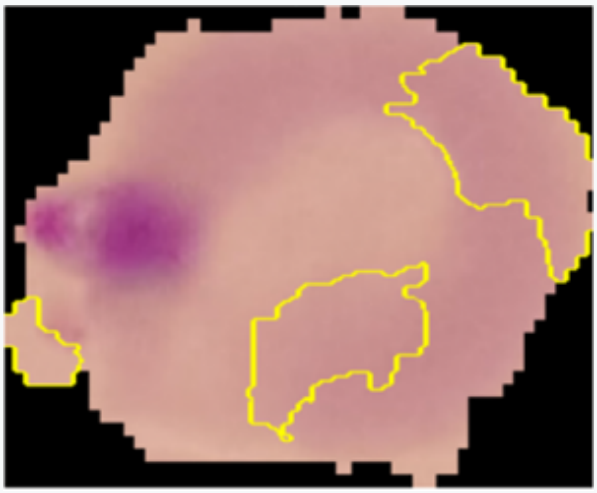
Difference. Our map has a more compact, block-like high-activation region near the right edge, with a transition extending inward. The paper's Xception example appears broader and more curved, spreading diagonally from the centre toward the right. The detailed localization and intensity distribution are not identical.

What this means. The outputs share a general spatial focus, but this is not evidence that the maps reproduce each other. Input image, checkpoint, preprocessing, target score, selected convolutional layer and heatmap normalization can all affect the result.

5. LIME: different superpixel selections



Our implementation



Original paper · Xception

Similarity. Both explanations use superpixel boundaries to identify regions selected by a local surrogate explanation.

Difference. Our output outlines a large, connected region across much of the central and right portion of the cell. The paper's Xception example selects a more separated pattern of regions, including upper-right and lower-central areas. The selected segments are visibly different.

What this means. LIME's segmentation and selected regions depend on the superpixel algorithm, segmentation settings and perturbation samples used to fit the local surrogate. A different pattern is not, by itself, proof of an implementation error or a better explanation.

Fair-comparison condition: The supplied material does not establish that the paper and our outputs use the same exact cell image, model checkpoint, preprocessing, target class and explanation parameters. We therefore report a qualitative comparison—not exact replication.

OUR EXPERIMENTS · QUANTITATIVE CHECKS

What we measured in our pilot

These values come from our project's supplied experiment outputs. They describe selected checks on a limited pilot, not a full clinical or dataset-level validation.

6. Pilot summary

Check	Our result	Interpretation
Classification pilot	20/20 correct; 10 images per class	Exploratory 20-image sample only; not full-dataset accuracy or independent holdout evidence.
LIME local fit	40 explanations; 1,000 samples each Mean R^2 = 0.4797; SD = 0.2155 Range = 0.2134–0.9229	Measures how well the local surrogate fits sampled predictions; it does not measure biological correctness.
SHAP additivity	19-row batch summary Mean absolute residual = 0.0720 Maximum = 0.1902	The residual describes additivity mismatch in this run; the twentieth case was handled separately as a diagnostic.
Grad-CAM stability	5 images; predictions unchanged Brightness SSIM = 0.9939; Spearman = 0.9934 Contrast SSIM = 0.9942; Spearman = 0.9963	High similarity under mild +10% brightness/contrast changes, but only for five images and these transformations.

7. Deletion-based faithfulness check

The deletion experiment progressively masks the most influential regions and records the predicted-class probability. Lower area under the deletion curve (AUC) indicates a faster probability drop under this setup. The comparison included Grad-CAM, SHAP and a random-order baseline; LIME was not included.

Method	Mean deletion AUC	SD	n
Grad-CAM	0.1419	0.0794	20
Random baseline	0.1247	0.1040	20
SHAP	0.1773	0.0955	20

Paired one-sided Wilcoxon tests did not show either attribution method significantly outperforming random ordering at the 0.05 level: Grad-CAM vs random p = 0.3506; SHAP vs random p = 0.0825.

Interpretation: This pilot does not provide statistically significant evidence that Grad-CAM or SHAP deletion ordering is more faithful than random ordering. Non-monotonic curves and masking artefacts are possible. The Grad-CAM ordering was derived from a heatmap PNG converted to grayscale, which may affect ordering if the saved heatmap uses a colour map.

Experimental setup notes

The deletion test used 196 patches of 16×16 pixels on 224×224 images, deletion fractions from 0 to 0.5, and a fill value of 0.5. Grad-CAM ordering used the saved heatmap converted to grayscale; SHAP contributions were oriented toward the predicted class. These implementation choices should be retained when interpreting or reproducing the results.

PHASE 2 EXTENSION · REPORTING LAYER

From explanation outputs to evidence-aware reports

Our Phase 2 contribution is a deterministic post-processing layer. It does not retrain the classifier or change its prediction. It reads the cross-method results and generates a report that makes available evidence and known limitations explicit.

8. Proposed workflow

XAI outputs Grad-CAM · LIME · SHAP	**Cross-method CSV** Common comparison record	**Evidence checks** Method-specific checks	**Natural-language report** Evidence + limitations

The reporting layer checks items such as Grad-CAM provenance/validity, LIME surrogate R^2 and area fractions, SHAP additivity residual, and limitations such as the absence of expert masks and the pilot scope. The SHAP residual threshold of 0.05 is a provisional project heuristic, not a universal scientific threshold.

9. Before-and-after reporting coverage

The evaluation counts whether predefined evidence and limitation fields are present in the generated reports. It measures field coverage—not whether each statement is factually correct or clinically useful.

Coverage measure	Baseline report	Evidence-gated report	Change
Evidence-item coverage	55.56%	100%	+44.44 percentage points
Limitation-item coverage	20%	100%	+80 percentage points

What improved? In the evaluated report outputs, every predefined evidence and limitation field was included by the evidence-gated version. This supports the claim that the reporting layer improves completeness against the project's checklist. It does not establish expert factuality, explanation faithfulness, clinician comprehension, or a reduced unsupported-claim rate.

10. How this addresses the research gap

Grad-CAM, LIME and SHAP produce different forms of explanation, and their outputs are not automatically interchangeable or direct proof of biological evidence. Our extension addresses the reporting side of this problem: instead of presenting an attribution visualization alone, it records method-specific checks and states when evidence is unavailable or evaluation is limited.

The contribution is therefore an evidence-aware reporting workflow, not a new classifier and not a new XAI algorithm. Phase 2 reporting evaluation was performed with Model 1 (Xception); the evidence-gated extension has not been demonstrated as evaluated on Model 2 in the supplied results.

INTERPRETATION & LIMITATIONS

What we can responsibly conclude

11. Overall findings

Area	Finding
Published baseline	The reference paper reports high classification accuracies for Xception and Inception-ResNetV2 on Kaggle and Harvard Dataverse datasets. These are the authors' results.
Our implementation	Our supplied outputs demonstrate that the project generates Grad-CAM, LIME and SHAP explanations for the input image.
Visual comparison	Our Grad-CAM has a broadly similar right-side focus to the paper's Xception example, but the detailed map shape differs. Our LIME superpixel pattern is visibly different from the paper's example.
Quantitative checks	The pilot provides selected local-fit, additivity, stability and deletion results. The deletion tests did not show statistically significant improvement over random ordering.
Phase 2 reporting	Evidence and limitation field coverage increased from 55.56% to 100% and from 20% to 100%, respectively, under the predefined checklist.

12. Limitations and scope

- The classification pilot contains only 20 images and should not be interpreted as full-dataset performance.
- The visual comparisons are representative examples, not controlled same-image reproductions.
- Grad-CAM stability was tested on five images with mild brightness and contrast changes only.
- LIME's local R^2 measures surrogate fit, not biological plausibility.
- SHAP additivity residual is a numerical diagnostic, not a pixel-level correctness measure.
- The deletion experiment did not show significant improvement over random ordering; LIME was not evaluated in that test.
- No expert parasite masks or clinician/user study were available; plausibility and usefulness were not evaluated.
- The reporting coverage evaluation measures checklist-field presence, not expert-verified factual accuracy.
- The Phase 2 evidence-gated reporting extension was evaluated with Model 1 (Xception), not Model 2.

13. Conclusion

This project extends a malaria XAI pipeline with a deterministic evidence-gated reporting layer. The implementation produces multiple explanation types, while the pilot experiments examine selected properties of those explanations. The clearest measured Phase 2 outcome is improved completeness of evidence and limitation reporting against a predefined checklist. Further work should use larger, controlled image sets, matched paper inputs and settings, expert annotations, and user evaluation before making stronger claims about explanation quality or clinical usefulness.

14. References

1. Parveen et al. (2025). “Trustworthy deep learning for malaria diagnosis using explainable artificial intelligence.” *Scientific Reports*, 15, 45037. https://doi.org/10.1038/s41598-025-28387-7

2. Lago, M. A., Zamzmi, G., Eich, B., & Delfino, J. G. (2026). “Evaluating Explainability: A Framework for Systematic Assessment of Explainable AI Features in Medical Imaging.” *Bioengineering*, 13(1), 111. https://doi.org/10.3390/bioengineering13010111

3. Ribeiro, M. T., Singh, S., & Guestrin, C. (2016). “Why Should I Trust You?” Explaining the Predictions of Any Classifier. KDD. https://doi.org/10.1145/2939672.2939778

4. Lundberg, S. M., & Lee, S.-I. (2017). “A Unified Approach to Interpreting Model Predictions.” NeurIPS.

5. Selvaraju, R. R. et al. (2017). “Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization.” ICCV. https://doi.org/10.1109/ICCV.2017.74

6. Dasgupta et al. (2022). “Framework for Evaluating Faithfulness of Local Explanations.” PMLR 162, 4794–4815. https://proceedings.mlr.press/v162/4794.html